

Project

American FX option with a Stochastic Interest Rate

Financial Engineering

Consider the European Option Market on the EUR-USD foreign exchange for an USD investor on the 8th of April 2016. Consider in particular the discount factors, the forwards (Mid Curve) and the ATM Black volatility at option expiry date $T=2y$ as reported in the file FXVolEURUSD.xlsm.

- i) Given an underlying dynamics with a stochastic IR, deduce the relations of the notes in attach for the Discount Factor and the Forward rate. Given Market conditions on the 8th of April 2016 for the EUR-USD market, calibrate the model for a 2y option. For the underlying volatility consider volatility at option expiry date $T=2y$ as quoted in the market, for the parameters σ_r, κ consider respectively the values 1% and 10%.
- ii) Implement the CRR tree approach described in [1] and [2]. Price American put options with expiry $T=2y$, $\rho = 0.3$, strikes between 80% S_0 and 120% S_0 . Plot the prices in function of the strike. For ATM-spot options plot the prices in function of ρ , $-1 < \rho < 1$. For ATM options, considering different sets of N values, discuss precision issues vs the computational time needed.
- iii) Compute the analytical solution for the corresponding European option and show that it has a Black-like form. Consider the case with $\rho = 0.3$, σ_r, κ as in point i), and show that the FX-volatility that should be used in the model with stochastic IR, in order to match FX option market prices, is different from the one used in i). Recompute American options with volatility value for which ATM European option is equal to market value (Black formula with market volatility). Compare American option results and European ones in the cases where there is a parallel shift in the Cost-of-carry (i.e. the net cost of holding positions in the underlying asset as defined in the document) of -2%, -1%, 1%, 2%.

Realize a library in Matlab.

Hints:

- i) Consider as random variable for interest rate only the OU process with zero mean-reverting value.
- ii) It is suggested to write the value of $\{S_{i+1}, r_{i+1}\}$ as a function $\{S_i, r_i\}$ and use this relation in Tree construction.
- iii) In order to find the solution for the corresponding European option with expiry T, consider the dynamics in the T-forward measure.

[1] J. Z. Wei, Valuing American Equity option with a Stochastic Interest Rate: A Note, The Journal of Financial Engineering, 2 (1996), 195-206.

[2] E. Appolloni, L. Caramellino and A. Zanette, A robust tree method for pricing American options with the Cox–Ingersoll–Ross interest rate model, IMA J. of Management Mathematics, (2014) 1-25.

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